

9. Write a Pandas program to create a Pivot table and find the total sale amount region-wise, manager-wise, and salesman-wise.

Aim

To create a Pivot Table and find the total sale amount region-wise, manager-wise, and salesman-wise using Pandas.

Procedure

1. Import the pandas library.
2. Read the sales data from a CSV file.
3. Create a Pivot Table.
4. Calculate the total sale amount.
5. Display the result.

Program and output:

```
import pandas as pd
from google.colab import files
uploaded = files.upload()
df = pd.read_csv("sales_region.csv")

pivot = pd.pivot_table(df,
                        values="Sale_Amount",
                        index=["Region", "Manager", "Salesman"],
                        aggfunc="sum")

print(pivot)
```

Choose Files sales_region.csv
sales_region.csv(text/csv) - 234 bytes, last modified: 7/23/2026 - 100% done
Saving sales_region.csv to sales_region.csv

			Sale_Amount
Region	Manager	Salesman	
East	Suresh	Deepak	35000
		Rani	27000
North	Ravi	Arun	25000
		Karan	30000
South	Meena	Priya	28000
		Rahul	22000
West	Anita	David	26000
		John	32000

Result

Thus, the Pivot Table showing the total sale amount region-wise, manager-wise, and salesman-wise was created successfully.

10. Create a DataFrame of ten rows and four columns with random values. Highlight negative numbers in red and positive numbers in black.

Aim

To create a DataFrame with random values and highlight negative numbers in red and positive numbers in black.

Procedure

1. Import NumPy and Pandas.
2. Generate a DataFrame with random values.
3. Define a function to apply colors.
4. Apply styling to the DataFrame.
5. Display the styled DataFrame.

Program and output:

```
import pandas as pd
import numpy as np

# Create random DataFrame
df = pd.DataFrame(np.random.randint(-50, 50, size=(10, 4)),
                  columns=["A", "B", "C", "D"])

# Function to color values
def color_number(value):
    if value < 0:
        return "color: red"
    else:
        return "color: black"

# Display styled DataFrame
print(df)

styled_df = df.style.map(color_number)

styled_df
```

```
...      A   B   C   D
0  44  20  21 -27
1  24 -31 -22 -49
2 -16 -12  36  17
3 -30 -33 -40 -14
4 -20  -5 -13  33
5 -27 -12  -3 -28
6 -40  -8 -50  19
7   9 -47  43 -39
8 -18  44  30  26
9 -13  48  10 -14
```

	A	B	C	D
...	0	44	20	21
				-27
	1	24	-31	-22
				-49
	2	-16	-12	36
				17
	3	-30	-33	-40
				-14
	4	-20	-5	-13
				33
	5	-27	-12	-3
				-28
	6	-40	-8	-50
				19
	7	9	-47	43
				-39
	8	-18	44	30
				26
	9	-13	48	10
				-14

Result

Thus, the DataFrame was created successfully, with negative numbers highlighted in red and positive numbers in black.